

State Variable Analysis

The differential equations of a lumped linear network can be written in the form

$$\begin{aligned}\dot{\mathbf{x}}(t) &= \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \\ \mathbf{y}(t) &= \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t)\end{aligned}\tag{1}$$

This system of first-order differential equations is known as the *state equation* of the system and $\mathbf{x}(t)$ is the state vector and $\mathbf{u}(t)$ is the input vector. The second equation is referred to as the *output equation*. $\mathbf{A}$ is called the state matrix, $\mathbf{B}$ the input matrix, $\mathbf{C}$ the output matrix, and $\mathbf{D}$ the direct transition matrix. One advantage of the state-space method is that the form lends itself easily to the digital and/or analog computer methods of solution. Further, the state-space method can be easily extended to analysis of nonlinear systems. State equations may be obtained from an n th-order differential equation or directly from the system model by identifying appropriate state variables.

To illustrate how we select a set of state variables, consider an n th-order linear plant model described by the differential equation

$$\frac{d^n y}{dt^n} + a_{n-1} \frac{d^{n-1} y}{dt^{n-1}} + \cdots + a_1 \frac{dy}{dt} + a_0 y = u(t)\tag{2}$$

where $y(t)$ is the plant output and $u(t)$ is the plant input. A state model for this system is not unique but depends on the choice of a set of state variables. A useful set of state variables, referred to as *phase variables*, is defined as

$$x_1 = y, \quad x_2 = \dot{y}, \quad x_3 = \ddot{y}, \quad \cdots, \quad x_n = y^{(n-1)}$$

Taking the derivatives, we have

$$\begin{aligned}\dot{x}_1 &= x_2, \quad \dot{x}_2 = x_3, \quad \dot{x}_3 = x_4, \quad \cdots, \quad \text{and } \dot{x}_n \text{ is given by (2).} \\ \dot{x}_n &= -a_0 x_1 - a_1 x_2 - \cdots - a_{n-1} x_n + u(t)\end{aligned}\tag{3}$$

or in matrix form

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \vdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} u(t)\tag{4}$$

and the output equation is

$$y = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \end{bmatrix} \mathbf{x}\tag{5}$$

Example 1

Obtain the state equation for the following differential equation.

$$2\frac{d^3y}{dt^3} + 4\frac{d^2y}{dt^2} + 6\frac{dy}{dt} + 8y = 10u(t)$$

The differential equation is third order, thus there are three state variables as follows

$$x_1 = y, \quad x_2 = \dot{y}, \quad x_3 = \ddot{y}$$

and the derivatives are

$$\dot{x}_1 = x_2, \quad \dot{x}_2 = x_3, \quad \text{and} \quad \dot{x}_3 = -4x_1 - 3x_2 - 2x_3 + 5u(t)$$

Or in matrix form

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -4 & -3 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} 5u(t)$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Simulation Diagram

Remember the simulation of differential equations which we studied earlier. Equation (3) indicates that state variables are determined by integrating the corresponding state equation. A diagram known as the *simulation diagram* can be constructed to model the given differential equations. The basic element of the simulation diagram is the integrator. The first equation in (3) is

$$\dot{x}_1 = x_2 \quad \text{Integrating, we have} \quad x_1 = \int x_2 dx$$

The above integral is represented by the time-domain diagram shown in Figure 1 similar to the block diagram.

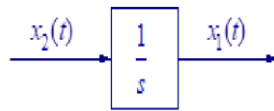


Figure 1 Simulation diagram for integrator

It is important to know that although the symbol $1/s$ is used for integration, the simulation diagram is a time domain representation. The number of integrators is equal to the number of state variables. For example, for the state equation in Example 1 we have three integrators in cascade, the three state variables are assigned to the output of each integrator as shown in Figure 2. The last equation in (3) is represented via a summing point and feedback paths.

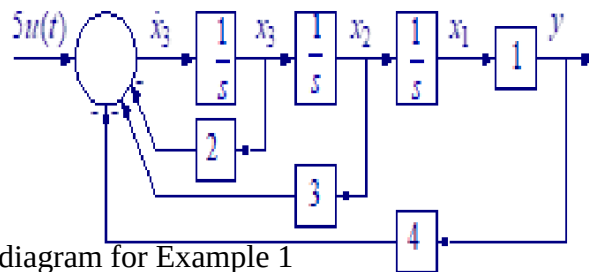


Figure 2 Simulation diagram for Example 1

Example 2

Consider the transfer function of a third-order system

$$\frac{Y(s)}{U(s)} = \frac{b_2 s^2 + b_1 s + b_0}{s^3 + a_2 s^2 + a_1 s + a_0} \quad (6)$$

where the numerator degree is lower than that of the denominator. The above transfer function is decomposed into two blocks as shown in Figure 3.



Figure 3 Transfer function (6) arranged in cascade form

For the first part (Figure 4), the state equations can be written as in the example 1. These equations will be the state equations for the complete transfer function shown in Figure 3.

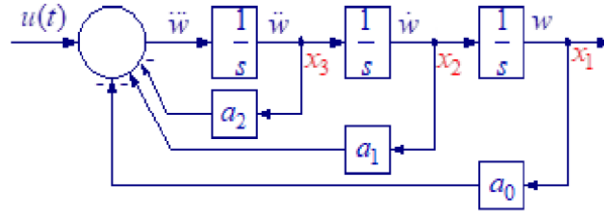


Figure 4

The state equations will be

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = x_3$$

$$\dot{x}_3 = -a_0 x_1 - a_1 x_2 - a_2 x_3 + u(t)$$

The only change is in the output equations. See the second part of the transfer function in the whole simulation diagram in Figure 5

Denoting the output of the first block as $W(s)$, we have

$$W(s) = \frac{U(s)}{s^3 + a_2 s^2 + a_1 s + a_0} \quad \text{and} \quad Y(s) = b_2 s^2 W(s) + b_1 s W(s) + b_0 W(s)$$

or

$$s^3 W(s) = -a_2 s^2 W(s) - a_1 s W(s) - a_0 W(s) + U(s)$$

This results in the following time-domain equation

$$\ddot{w} = -a_2 \ddot{w} - a_1 \dot{w} - a_0 w + u(t) \quad \text{and} \quad y(t) = b_2 \ddot{w} + b_1 \dot{w} + b_0 w$$

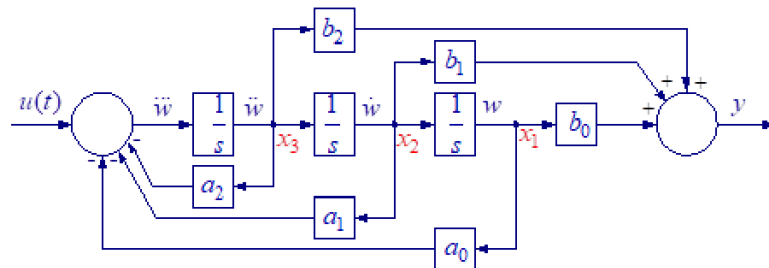


Figure 5 Simulation diagram of the complete transfer function

The output equation, which is in time domain, is

$$y = b_0 x_1 + b_1 x_2 + b_3 x_3$$

or in matrix form

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u(t) \quad (8)$$

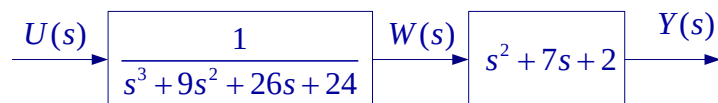
$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

The only difference from the first example is the change in output equations.

Example 3

For the following transfer function

$$G(s) = \frac{Y(s)}{U(s)} = \frac{s^2 + 7s + 2}{s^3 + 9s^2 + 26s + 24}$$



Check the solution is write or not

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = x_3$$

$$\dot{x}_3 = -24x_1 - 26x_2 - 9x_3 + u(t)$$

and the output equation is

$$y = 2x_1 + 7x_2 + x_3$$

or in matrix form

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -24 & -26 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u(t)$$

$$y = \begin{bmatrix} 2 & 7 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

State-Space to Transfer Function Conversion

Consider the state and output equations

$$\dot{x}(t) = \mathbf{A}x(t) + \mathbf{B}u(t)$$

$$y = \mathbf{C}\mathbf{x}(t) + \mathbf{D}u(t)$$

Taking the Laplace transform

$$sX(s) = AX(s) + BU(s) \Rightarrow [sI - A]X(s) = BU(s)$$

$$Y(s) = CX(s) + DU(s)$$

Substituting for $X(s)$ in the second equation above, we get

$$Y(s) = C[sI - A]^{-1}BU(s) + DU(s)$$

or

$$\frac{Y(s)}{U(s)} = C[sI - A]^{-1}B + D$$

Example 4

A system is described by the following state-space equations

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

$$y = [8 \quad 1] \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix}$$

Obtain the system transfer function using the formula in (3.18)

$$[sI - A] = \begin{bmatrix} s & -1 \\ 6 & s+5 \end{bmatrix} \Rightarrow \Phi(s) = [sI - A]^{-1} = \frac{\begin{bmatrix} s+5 & 1 \\ 6 & s \end{bmatrix}}{s^2 + 5s + 6}$$

$$G(s) = C[sI - A]^{-1}B = [8 \quad 1] \frac{\begin{bmatrix} s+5 & 1 \\ 6 & s \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}}{s^2 + 5s + 6} = \frac{[8 \quad 1] \begin{bmatrix} 1 \\ s \end{bmatrix}}{s^2 + 5s + 6} = \frac{8 + s}{s^2 + 5s + 6}$$

Therefore

$$G(s) = \frac{s+8}{s^2 + 5s + 6}$$

How to obtain state variable model for electric circuits

Later will be studying a systematic method for writing state equations for electric circuits.

For obtaining the state variable model of simple circuits, you can follow the steps steps:

1. Take inductor currents and capacitor voltages as state variables
2. Assign directions for the inductor currents and polarity of the capacitor voltages
3. Write KCL equations at the capacitor ends (cutsets)
4. Write KVL equations in the inductor loops (circuits)

Example 5

Write the state equation for the network shown in Figure 7.

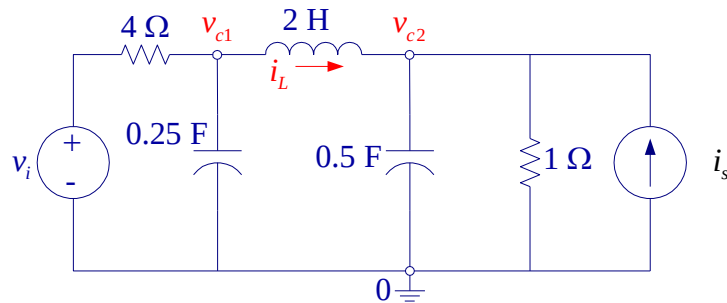


Figure 7 Circuit of Example 2

Define the state variables as current through the inductor and voltage across the capacitors. Write two node equations containing capacitors and a loop equation containing the inductor. The state variables are v_{c1} , v_{c2} , and i_L .

Node equations are

$$0.25 \frac{dv_{c1}}{dt} + i_L + \frac{v_{c1} - v_i}{4} = 0 \quad \Rightarrow \quad \dot{v}_c = -v_{c1} - 4i_L + v_i$$

$$0.5 \frac{dv_{c2}}{dt} - i_L + \frac{v_{c2}}{1} - i_s = 0 \quad \Rightarrow \quad \dot{v}_{c2} = 2i_L - 2v_{c2} + 2i_s$$

and the loop equation is

$$2 \frac{di_L}{dt} + v_{c2} - v_{c1} = 0 \quad \Rightarrow \quad \dot{i}_L = 0.5v_{c1} - 0.5v_{c2}$$

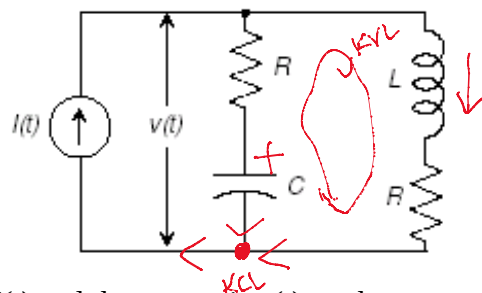
or

$$\begin{bmatrix} \dot{v}_{c1} \\ \dot{v}_{c2} \\ \dot{i}_L \end{bmatrix} = \begin{bmatrix} -1 & 0 & -4 \\ 0 & -2 & 2 \\ 0.5 & -0.5 & 0 \end{bmatrix} \begin{bmatrix} v_{c1} \\ v_{c2} \\ i_L \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_i \\ i_s \end{bmatrix}$$

See the order of the B matrix. There are 3 state variables and 2 inputs.

Example 6

Consider the electrical circuit shown below:



The input is $I(t)$ and the output is $v(t)$ as shown. Determine the transfer function of the system.

Obtain the state variable representation of the system. Obtain A, B, C and D matrices

(Assume capacitor voltage as x_1 and inductor current as x_2)

Obtain the transfer function of the system from A, B, C and D matrices.

Transfer function is $Z(s)$

$$Z(s) = \left(R + \frac{1}{Cs} \right) \text{ is in parallel with } R + Ls = \frac{RLCs^2 + (L + RC^2)s + R}{LCs^2 + 2RCs + 1}$$

Writing KCL at capacitor junction, $C\dot{x}_1 + x_2 - I(t) = 0$

$$\dot{x}_1 = \frac{-1}{C}x_2 + \frac{1}{C}I(t) \dots\dots\dots (1)$$

Writing KVL in the inductor loop, $L\dot{x}_2 + Rx_2 - x_1 + R(x_2 - I(t)) = 0$

$$x_2 = \frac{1}{L}x_1 - \frac{2R}{L}x_2 + \frac{R}{L}I(t) \dots\dots\dots (2)$$

Output equation is $V(t) = x_1 - Rx_2 \dots\dots\dots (3)$

From the above equations

$$A = \begin{bmatrix} 0 & \frac{-1}{C} \\ \frac{1}{L} & \frac{-2R}{L} \end{bmatrix} \quad B = \begin{bmatrix} \frac{1}{C} \\ \frac{R}{L} \end{bmatrix} \quad \text{and} \quad C = [1 \quad -R]$$

Example 7

Solve the state equation $\begin{matrix} \dot{x}_1 = x_2 \\ \dot{x}_2 = -2x_1 - 3x_2 + 1 \end{matrix} \quad X(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$$X(s) = (sI - A)^{-1}x(0) + (sI - A)^{-1}BU(s)$$

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad X(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad u = 1 \quad U(s) = \frac{1}{s}$$

$$(sI - A)^{-1}x(0) = \frac{1}{s^2 + 3s + 2} \begin{bmatrix} s + 3 \\ -2 \end{bmatrix}$$

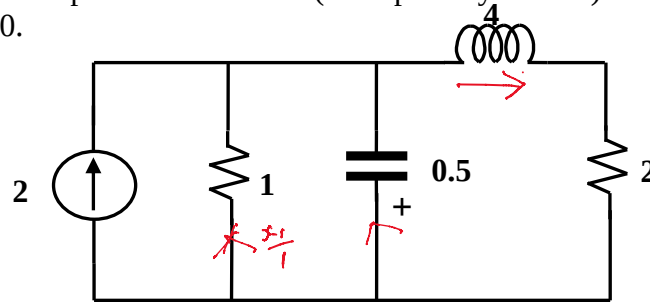
$$(sI - A)^{-1}BU(s) = \frac{1}{s^2 + 3s + 2} \begin{bmatrix} 1 \\ s \end{bmatrix} \frac{1}{s} \quad \text{Add these and find the inverse Laplace transform}$$

Finding inverse transform, $x_1(t) = 0.5 + e^{-t} - 0.5e^{-2t}$. **Complete the problem.**

Example 8

In the network shown in figure, the outputs are the voltages across the inductance and resistances. Derive a state space model and use this model **to obtain the outputs**. The

initial voltage across the capacitor is 1V (with polarity shown) and initial current through the inductor is 0.



Selecting capacitor voltage as x_1 (with polarity shown) and inductor current as x_2 (assuming the direction shown), KCL at A gives

$$0.5\dot{x}_1 + x_1 + 2 - x_2 = 0 \quad \text{i.e., } \dot{x}_1 = -2x_1 + 2x_2 - 4 \text{ ----- (1)}$$

Writing KVL,

$$x_1 + 4\dot{x}_2 + 2x_2 = 0 \quad \text{i.e., } \dot{x}_2 = -\frac{1}{4}x_1 - \frac{1}{2}x_2 \text{ (2)}$$

Let y_1 be the voltage across inductance, y_2 voltage across 1Ω resistor and y_3 voltage across 2Ω resistor.

$$y_1 = 4\dot{x}_2 \quad \text{i.e., } y_1 = -x_1 - 2x_2 \text{ (3)}$$

$$y_2 = x_1 \quad y_2 = x_1 \text{ (4)}$$

$$y_3 = 2x_2 \quad y_3 = 2x_2 \text{ (5)}$$

Equations (1) to (5) represent the state variable model

$$\text{Taking the Laplace transform of (1), } sX_1(s) - x_1(0) = -2X_1(s) + 2X_2(s) - \frac{4}{s} \text{ ----- (6)}$$

$$\text{Taking the Laplace transform of (2) } sX_2(s) - 0 = -\frac{1}{4}X_1(s) - \frac{1}{2}X_2(s).$$

$$\text{From the above equation, } X_1(s) = -(4s + 2)X_2(s) \text{ ----- (7)}$$

Substituting this value of $X_1(s)$ and $x_1(0) = 1$

$$-X_2(s)[4s^2 + 2s] - 1 = X_2(s)[8s + 4] + 2X_2(s) - \frac{4}{s}$$

$$\text{From which } X_2(s) = \frac{4-s}{s(4s^2 + 10s + 6)} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+1.5} \quad A = \frac{2}{3}, B = -2.5, C = \frac{11}{6}$$

$$x_2(t) = \frac{2}{3} - 2.5e^{-t} + \frac{11}{6}e^{-1.5t} \text{ (8)}$$

$$\dot{x}_2(t) = 2.5e^{-t} - \frac{33}{12}e^{-1.5t} \text{ ----- (9)}$$

From (7), $x_1(t) = -4\dot{x}_2(t) - 2x_2(t)$ Substituting the values of $x_2(t)$ and $\dot{x}_2(t)$ from (8) and (9)

$$x_1(t) = -\frac{4}{3} - 5e^{-t} + \frac{22}{3}e^{-1.5t} \text{ --- (10)}$$

Second method:

$$\mathbf{X}(s) = (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{x}(0) + (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}U(s)$$

$$\mathbf{A} = \begin{bmatrix} -2 & 2 \\ -0.25 & -0.5 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} -2 \\ 0 \end{bmatrix} \quad \mathbf{X}(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad u = 2 \quad U(s) = \frac{2}{s}$$

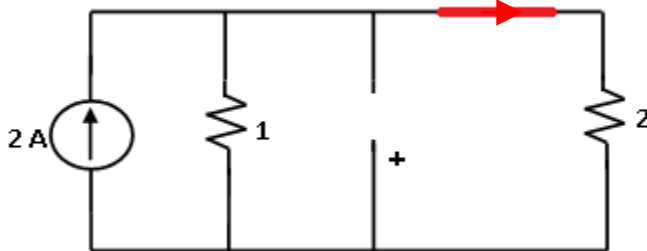
$$(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{x}(0) = \frac{1}{s^2 + 2.5s + 1.5} \begin{bmatrix} s + 0.5 \\ -0.25 \end{bmatrix}$$

$$(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}U(s) = \frac{1}{s^2 + 2.5s + 1.5} \begin{bmatrix} -2s - 1 \\ 0.5 \end{bmatrix} \frac{2}{s} \quad \text{You will be getting the same solution.}$$

The given circuit's behavior at $t = \infty$.

$$\text{Current through } 2\Omega \text{ resistor} = 2 \times \frac{1}{1+2} = \frac{2}{3} \text{ A} = x_2(\infty)$$

$$\text{Voltage across } 2\Omega \text{ resistor} = 2 \times \frac{2}{3} = \frac{4}{3} \text{ V} = x_1(\infty)$$



$$\mathbf{X}(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \mathbf{X}(\infty) = \begin{bmatrix} \frac{2}{3} \\ \frac{4}{3} \end{bmatrix}$$

Substitute the values of $x_1(t)$ and $x_2(t)$ from equations (8) and (10) at $t = 0$ and $t = \infty$ and check you are getting the correct $\mathbf{X}(0)$ and $\mathbf{X}(\infty)$.

Equilibrium points

The equilibrium point is in the state space at which the dynamical system will stay if it starts from that point. An equilibrium point is one where the state variables are constant and unchanging. Since the derivatives represent changes in the state variables, this statement is equivalent to saying the derivatives for the state variables are 0 at equilibrium points.

If the eigenvalues of the $\mathbf{A}$ matrix are in the LHS plane, then the equilibrium point is stable.

Equilibrium points of a linear systems are stable node, stable focus, unstable node, unstable focus, centre and saddle points.

If the eigenvalues are real and negative, then equilibrium point is a stable node. If the eigenvalues are real and positive it is unstable node.

Find the equilibrium points of the circuits describe by the following equations:

$$1. \quad \ddot{x} + \dot{x} + x = 0$$

$$2. \quad \ddot{x} + 0.5\dot{x} + 2x + x^2 = 0$$

$$1. \quad \begin{aligned} x_1 &= x \\ x_2 &= \dot{x} \\ \dot{x}_2 &= \ddot{x} \end{aligned}$$

$$\begin{aligned} x_1 &= x_2 \\ \dot{x}_2 &= -x_1 - x_2 \end{aligned}$$

stable focus

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}$$

eigenvalues
complex -ve
real part

Solve $0 = x_2$

$$0 = -x_1 - x_2$$

- Eqbm point is $(0,0)$ is a stable focus

2. The second equation is nonlinear.

$$\ddot{x} + 0.5\dot{x} + 2x + x^2 = 0 \quad \text{Non linear}$$
$$\begin{aligned} x_1 &= x \\ x_2 &= \dot{x} \end{aligned} \quad \begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -2x_1 - 0.5x_2 - x_1^2 \end{aligned}$$
$$x_2 = 0 \quad x_1 = 0 \quad \begin{aligned} 2x_1 &= -x_1^2 \\ x_1 &= -2 \end{aligned}$$

$(0,0)$ $(-2,0)$ are the equilibrium points

```
function c = non(t,x);  
c = [x(2); -2*x(1) - 0.5*x(2) - (x(1)*x(1))];
```

save this as a file non.m

```
ts = [0 20];  
x0 = [0.2; 0.1];  
[t,x] = ode23('non', ts, x0);  
plot(x(:,1), x(:,2))
```

Type this in the command window

